Application-Sector-Wise Production Profile of the Indian Electronics Sector

Item/Vertical	2014-15 (₹ billion)	2015-16 (₹ billion)	2016-17 (₹ billion)	2017-18 (₹ billion)	2018-19 (₹ billion)
Consumer electronics	558.06	557.65	647.42	735.24	770.00
Industrial electronics	393.74	450.83	622.14	690.57	808.50
Computer hardware	186.91	198.85	203.82	214.01	211.80
Cellphones	189.00	540.00	900.00	1320.00	1700.00
Strategic electronics	157.00	180.55	207.60	235.62	282.7
Light emitting diodes (LEDs)	21.72	50.92	71.34	96.30	130.00

Note: Data above is as provided by the respective industry associations including CEAMA, ELCINA, MAIT and ICEA. (Source: MeitY Annual Report, 2018-19)

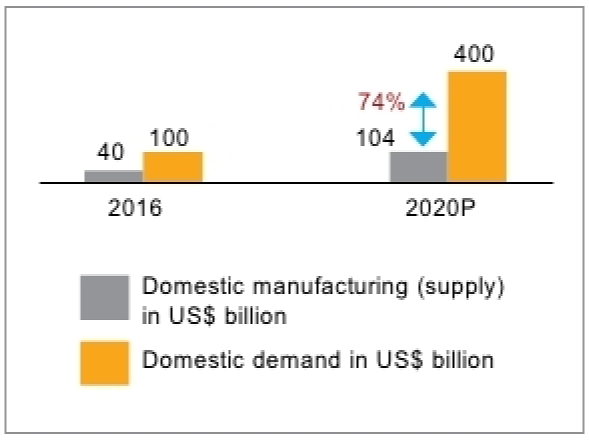

Fig. 2: Demand versus supply (domestic manufacturing) gap in the Indian electronics system design and manufacturing (ESDM) sector (Source: IBEF, MeitY)

Local brands under Make in India initiative as well as global manufacturers looking to relocate their manufacturing base from China to such alternate locations as India, Vietnam and Indonesia (due to mounting labour costs in China) are driving the trend. Thus, India has the potential to emerge as a global EMS hub.

Market at a glance

According to a ResearchAndMarkets report, the Indian EMS market is expected to grow at a CAGR of 38.1 per cent till FY23 from the market value of approximately ₹ 560 billion in 2018. The EMS market in India will be highly dynamic in the coming year since India promises to emerge as a hotspot for electronics manufacturing among south-Asian nations. For this the Indian government is providing several incentives.

The ever-increasing demand for cellphones, telecom products, and consumer and smart electronic devices is driving the growth of the EMS market (as shown in the table). Manufacturing partnerships in these segments have been growing steadily, as market demand for the products is extremely high and OEMs are striving to cut costs to maintain their competitive advantage in the face of rapidly-changing market conditions, technological advances and global competition.

There are further opportunities for EMS providers outside these segments, namely, in lighting, power electronics, automotive and strategic electronics (including aerospace, defence and railways). With increasing demand for the transition from electrical to electronic in the energy segment, power electronics is emerging as one of the most exciting areas in the industry. Aerospace and defence (A&D) OEMs have been increasingly depending on EMS providers to address risk management, logistics and aftermarket service needs. EMS companies with global supply chains and advanced technological capabilities are in the best place to exploit this trend.

Moreover, rapidly-evolving devices like smartphones and tablets account for the development of unique collaborations and partnerships between service providers and OEMs. This, in turn, directs EMS market players to deploy efficient manufacturing machines and service delivery facilities, which further fuels industry development.

Additionally, increasing investments in outsourcing activities by OEMs for availing logistics, manufacturing and testing services add up to the consumption of products.

Demand for green and energy-efficient devices and components that creates a need for these services further contributes to the growth of the EMS market. To cater to these demands, OEMs are developing strategic partnerships with service providers that aid them in controlling manufacturing costs. Service providers offer product design, development and manufacturing operations according to demand from OEMs. Due to these facilities, OEMs are able to achieve appropriate product supplies and save on major development costs. This trend allows EMS market players to develop green manufacturing facilities and energy-efficient products that support globally sustainable goals.

Indian EMS providers tend to show long-term affinity towards the high-volume, low-mix (HVLM) model as it ensures economies of scale, higher productivity due to minimum change-over time and optimum machine utilisation. Considering the low margin and highly competitive price scenario, optimisation of supply chain cost is crucial to the success of this model.

However, recent industry trends show a shift towards high-mix, low-volume (HMLV) model, which keeps high focus on quality and customisation as per customer requirements. Considering the high margin and niche market scenario, even major changes in market dynamics often do not impact such a production process heavily. However, in this model success depends on controlling and improving supply chain efficiently.

Innovative services in demand

In response to growing competition in the industry, EMS providers continuously adopt innovative strategies. These include better knowledge of customer needs, understanding business models of customers/OEMs, effective communication tools, creating global footprints, focusing on core competencies and most effectively offering value-added services. Moreover, considering intense competition, they need to enhance their value proposition by offering integrated and end-to-end solutions, and other innovative services, as given below.